

# Object on Use Case Description: Sequence Diagram Conformance based on Step Performed using Text Pre-Processing on Sipranta Application SRS

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**Abstract**—In the Use Case Diagram, a Step Performed in Tabulation contains an object's explanation flow. A sequence diagram attempts to depict in detail the interaction of objects in a system, the messages or commands sent, and the implementation time. In other words, the Sequence Diagram, like the Use Case Diagram, encapsulates an object's explanation flow in a graphical diagram. The Step Performed and the Sequence Diagram should be related, yet the Step Performed and Sequence Diagram diverge while creating the Sipranta Application SRS. By adopting Text Pre-processing, this research intends to quantify the value of conformance between the Sequence Diagram and the Step Performed in Use Case Description. This study identifies the Step Performed and the Sequence Diagram using 22 documents, 11 of which are Step Performed and 11 of which are Sequence Diagrams. In addition, the Text Analysis process produces two documents with the highest conformity: d1 with d12 of 0.933461. Cohen Cappa Score produces a validation value of 0.29195, which is included in the "Fair Agreement" category. While the reliability results from 2 groups of experts, which is 0.8275, so it is included in the "Substantial agreement" category. The results differ between the two methods, which is 0.536.

**Keywords**— Sequence Diagram, Step Performed, Text Pre-Processing.

## I. INTRODUCTION

In general, in the development of software or software, a document is needed that can explain as a whole about the software such as the features in the software, the logic flow of the software, and the hardware specifications used during the development of the software and software, the document is called Software Requirement Specification (SRS) [1, 2, 3]. Sipranta is an Android application that aims to assist in implementing activities at posyandu. Posyandu is a center for pre and postnatal health services and information for women and children under five years old. Among the features of Sipranta are toddler data management, toddler immunization data management, examination schedules, and others. Fig. 1 depicts the use case modelling of the Sipranta application.

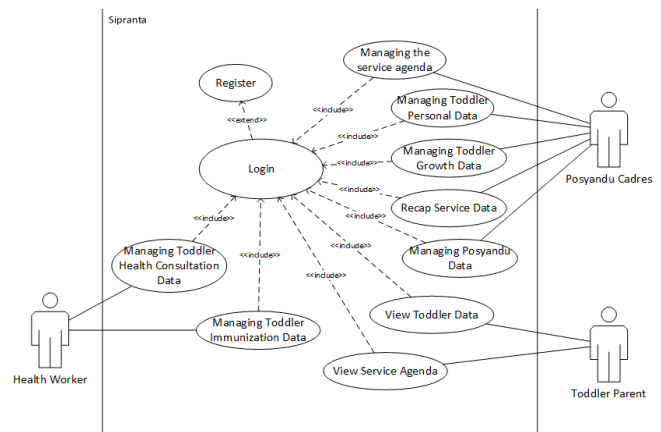


Fig. 1. Use Case Modelling of Sipranta

The SRS document must be created appropriately under the Use Requirements for the program to be constructed as requested [4, 5]. A use case diagram, class diagram, sequence diagram, user interface, activity diagram and other artefacts are included in the SRS document. These artefacts must be interconnected to make the program developed clearer than required in software development.

Unified Modeling Language (UML) has a use case diagram to depict the interaction connection between actors and the system [4, 6]. On the other hand, the Use Case Description has Step Performed in the form of tabulation, which comprises an object's explanation flow in the Use Case Diagram. A sequence diagram attempts to depict the interaction of objects in a system, the messages or commands conveyed, and the implementation time. In other words, the Sequence Diagram, like the Use Case Diagram, encapsulates an object's explanation flow in a pictorial diagram [7].

The Sequence Diagram and the Step Performed should be linked. However, when developing the Sipranta Application SRS, the Step Performed and the Sequence Diagram have differences, so to overcome this problem, similarity measurements must be made between the two artifacts by implementing Text Pre-processing, which previously extracted the two artifacts. An assessment of the quality of the SRS

created can be made by measuring the similarity between the two artifacts.

By adopting Text Pre-processing, this study intends to quantify the value of conformance between the Sequence Diagram and the Step Performed. Contributors to this study include: The goal is to quantify the value of conformance between the Sequence Diagram and the Step Performed section in Use Case Description using Text Pre-processing. This study makes the following contributions:

- Perform Text Pre-processing on the two extracted artifacts.
- Perform conformity measurements using the cosine similarity method from the results of Text Pre-processing.
- Perform validity and reliability evaluations on the findings of conformity assessment between Step Performed Section in Use Case Description and Sequence Diagrams.
- Generate a Sequence Diagram appropriateness value based on the Step Performed in the Sipranta SRS, which generates objects to be compared with the Use Case Description.

## II. RELATED WORK

### A. Software Requirements Specification

The Software Requirements Specification (SRS) document must be completed to develop a software system successfully. It provides a description of the requirements as well as detailed software specifications [8]. This SRS document serves to guide all parties involved in software development. With the SRS, software development can be more comfortable [1, 2]. This SRS can function or not function, depending on the needs. The SRS document also contains the Unified Modelling Language (UML) [8].

### B. Use Case Description (Step Performed)

Use Case Description is an SRS component that descriptively explains a Use Case in a format of business process functionality that includes creating a system [9, 10]. Use Case Description is usually made in tabular form. Use Case Description has several components: Use Case name, Scenario, Actor, and Step Performed.

### C. Sequence Diagram

Sequence Diagrams are UML artifacts in SRS that model aspects of system behavior in detail [11, 12]. Object-oriented software is very suitable for this Sequence Diagram, where the Sequence Diagram visualizes control flow during object interaction. Sequence Diagrams contain a set or sets of objects that interact with each other in sequence and exchange messages between objects. Sequence Diagrams start by selecting the activity to be executed first, running all processes, and displaying what changes occur to the input or output.

### D. Text Pre-Processing

Text pre-processing is extracting information automatically from various sources in writing or text to find new information previously unknown to the computer. The goal is to connect the retrieved information to produce new facts or hypotheses that may be investigated further using more traditional experimental methods [13, 14, 15].

Text pre-processing is a component of the text mining approach. Text mining is a subset of data mining that seeks intriguing patterns in huge data sets [16, 17]. Regular data mining differs from text mining in that text mining extracts patterns from natural language text, whereas data mining extracts patterns from a structured database [13, 18].

### E. Cosine Similarity

Cosine Similarity is a frequently used statistic for information retrieval. The basic idea behind this approach is to represent text as a vector of phrases [19]. Cosine similarity computes the similarity between two documents by computing the cosine value between the vectors of the two modelled papers [20, 21]. This measure may be used for two types of texts (documents, paragraphs, or phrases).

One case commonly found in the use of cosine similarity is the case of search engines, where user requests will be calculated with stored documents, and documents with the highest value will be ranked at the top as recommendations for users [19]. Cosine similarity can be seen in formula (1).

$$\text{similarity} = \cos(\theta) = \frac{A \cdot B}{\|A\| \|B\|} = \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2} \sqrt{\sum_{i=1}^n B_i^2}} \quad (1)$$

### F. Gwet's AC1

Gwet's AC1 is a calculating approach based on a model agreement between two experts [9, 22]; the result of the agreement is an index result with several categories, as shown in Table I. Gwet's AC1 can be used to examine validity and reliability.

TABLE I. GWET'S VALUE INDEX AC1

| Kappa Index | Proportion Agreement       |
|-------------|----------------------------|
| < 0.00      | Less than chance-agreement |
| 0.01 – 0.20 | Slight agreement           |
| 0.21 – 0.40 | Fair agreement             |
| 0.41 – 0.60 | Moderate agreement         |
| 0.61 – 0.80 | Substantial agreement      |
| 0.81 – 1.00 | Almost perfect             |

The steps for implementing Gwet's AC1 are as follows:

1) Using the method in Table II, create a table with the findings of the needs elicitation between the two experts.

TABLE II. ELICITATION RESULTS BETWEEN TWO EXPERTS

| Expert 1 | Expert 2 |          |          |
|----------|----------|----------|----------|
|          | Yes      | No       | Total    |
| Yes      | A        | B        | B1 = A+B |
| No       | C        | D        | B2 = A+C |
| Total    | A1 = A+C | A2 = B+D | N        |

2) Using formula (2) to calculate the observed agreement (P)

$$P = \frac{A+D}{N} \quad (2)$$

3) Probabilitas Calculator Using formula (3), Change-agreement or e(Y)

$$e(Y) = 2P_1(1-P_1) \quad (3)$$

Where for  $P_1$ :

$$P_1 = \frac{A_1 + B_1}{N} \quad (4)$$

Information:

$A_1$  and  $B_1$  expert numbers 1 and 2, respectively, and  $N$  is the quantity of data utilized.

4) Data is categorized as "Yes" depending on the findings of the elicitation table.

5) Calculating AC1 statistics with the formula Agreement Coefficient.

$$AC1 = \frac{P-e(Y)}{1-e(Y)} \quad (5)$$

### III. DATASETS

Datasets are created by recognizing SRS artifacts such as the Step Performed and the Sequence Diagram. A sentence will be constructed using the text that was extracted from the artifacts. Each the extracted Step Performed and Sequence Diagram will be labeled as a document. Table III illustrates the labeling of each Use Case in the Step Performed and the indicated Sequence Diagram.

TABLE III. DOCUMENT LABELING FROM THE IDENTIFICATION RESULTS OF STEP PERFORMED AND SEQUENCE DIAGRAM

| No | Use Case Name                             | Step Performed |                | Sequence Diagram |                |
|----|-------------------------------------------|----------------|----------------|------------------|----------------|
|    |                                           | ID             | Document Label | ID               | Document Label |
| 1  | Register                                  | SP01           | d1             | SD01             | d12            |
| 2  | Login                                     | SP02           | d2             | SD02             | d13            |
| 3  | Managing the Service Agenda               | SP03           | d3             | SD03             | d14            |
| 4  | Managing Toddler Personal Data            | SP04           | d4             | SD04             | d15            |
| 5  | Managing Toddler Growth Data              | SP05           | d5             | SD05             | d16            |
| 6  | Recap Service Data                        | SP06           | d6             | SD06             | d17            |
| 7  | Managing Posyandu Data                    | SP07           | d7             | SD07             | d18            |
| 8  | Managing Toddler Immunization Data        | SP08           | d8             | SD08             | d19            |
| 9  | Managing Toddler Health Consultation Data | SP09           | d9             | SD09             | d20            |
| 10 | View Toddler Data                         | SP10           | d10            | SD10             | d21            |
| 11 | View Service Agenda                       | SP11           | d11            | SD11             | d22            |

### IV. METHODOLOGY

Referring to Fig. 2, this research generally implements the concepts of Text Pre-Processing, Similarity Measurement, Validity, and Reliability.

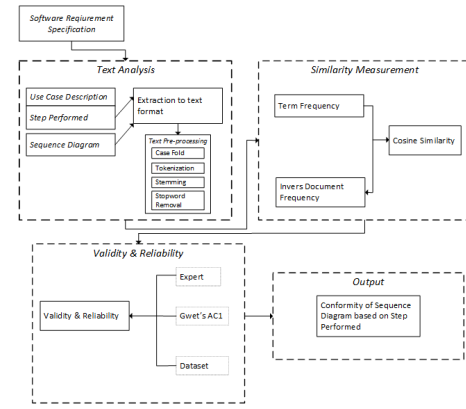


Fig. 2. Research activities

This study's output is an appropriateness score between sequence diagrams based on the steps performed. The suitability increases with increasing appropriateness value (near score=1) [19, 21, 23].

Fig. 2 is a diagram of the methodology used in this study. Several comparison parameters will be used, namely:

- There are 11 use cases, each containing a Step Performed and a Sequence Diagram, totaling 24 documents—11 from the Step Performed and 11 from the Sequence Diagram.
- Each document's text preprocessing uses Case Folding, Tokenization, Stemming, and Stopword Removal. [24].
- Word weighting in use using TF-IDF matrix.
- The measurement method used is Cosine Similarity.
- Similarity between two documents can be considered appropriate if it exceeds the number  $> 0.8$
- Validation and Reliability are done with Cohen's Kappa and Gwet's AC1 methods. The results of these two methods will be compared with the measurement results using Cosine Similarity. If the value is close, the measurement results are valid and reliable.

### V. RESULT AND DISCUSSION

This section will explain the steps based on the methodology that has been described, starting from Text Analysis to the resulting output.

#### A. Text Analysis

The first step in the Text Analysis process is to build a dataset by extracting SRS artifacts: Step Performed and Sequence Diagrams.

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Use case name  | Register                                                                                                                                                                                                                                                                                                                                                                                                                                        | Id: UC01 |
| Use case name  | Login                                                                                                                                                                                                                                                                                                                                                                                                                                           | Id: UC02 |
| Use case name  | Managing the Service Agenda                                                                                                                                                                                                                                                                                                                                                                                                                     | Id: UC03 |
| Use case name  | Managing Toddler Personal Data                                                                                                                                                                                                                                                                                                                                                                                                                  | Id: UC04 |
| Use case name  | Managing Toddler Growth Data                                                                                                                                                                                                                                                                                                                                                                                                                    | Id: UC05 |
| Use case name  | Recap Service Data                                                                                                                                                                                                                                                                                                                                                                                                                              | Id: UC06 |
| Actor          | Posyandu Cadres                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |
| Description    | Make a recap of data on posyandu activity services to become a reporting document to the puskesmas.<br>1. Posyandu cadres open the sipranta application<br>2. Posyandu cadres press the profile menu on the main page<br>3. Posyandu cadres press the data recap button on the profile page<br>4. Posyandu cadres select the data they want to recap on the data recap page<br>5. The sipranta application generates recap data in excel format |          |
| Step Performed | 5. The sipranta application generates recap data in excel format                                                                                                                                                                                                                                                                                                                                                                                |          |
| Preconditions  | Posyandu cadres have not recapulated service data                                                                                                                                                                                                                                                                                                                                                                                               |          |
| Postconditions | Posyandu cadres succeeded in recapulating service data                                                                                                                                                                                                                                                                                                                                                                                          |          |
| Assumptions    | Posyandu cadres already have an account and log in                                                                                                                                                                                                                                                                                                                                                                                              |          |
| Preconditions  | Posyandu cadres have succeeded in managing toddler growth data                                                                                                                                                                                                                                                                                                                                                                                  |          |
| Postconditions | Posyandu cadres already have an account and log in                                                                                                                                                                                                                                                                                                                                                                                              |          |
| Assumptions    | Posyandu cadres already have an account and log in                                                                                                                                                                                                                                                                                                                                                                                              |          |

Fig. 3. Snapshot of Step Performed on Sipranta Application SRS

The eleven Steps Performed used can be seen in Fig.3, which are four examples of datasets identification results. See Table IV

TABLE IV. IDENTIFICATION RESULTS OF STEP PERFORMED

| Document Label | Identification Results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| d1             | parents of toddlers, posyandu cadres, health workers open the sipranta application. parents of toddlers, posyandu cadres, health workers press the register button on the login page. parents of toddlers, posyandu cadres, health workers input register data on the register page. the sipranta application validates the register data. the sipranta application sends a verification email to parents of toddlers, posyandu cadres, health workers. parents of toddlers, posyandu cadres, health workers verify accounts. |
| d2             | parents of toddlers, posyandu cadres, health workers open the sipranta application. parents of toddlers, posyandu cadres, health workers input login data on the login page. the sipranta application validates login data. the sipranta application displays the main page.                                                                                                                                                                                                                                                  |
| d3             | posyandu cadres open the sipranta application. posyandu cadres press the service agenda button on the main page. posyandu cadres select management options on the service agenda page. posyandu cadres input the service agenda data on the service agenda form page. the sipranta application validates the service agenda data. the sipranta application stores service agenda data.                                                                                                                                        |
| ...            | ...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| d11            | toddler parents open the sipranta application. parents of toddlers press the service agenda list menu on the main page. the sipranta application displays a list of service agendas on the service agenda list page. parents of toddlers pressing service agenda items on the service agenda list page. the sipranta application displays details of the service agenda on the service agenda page.                                                                                                                           |

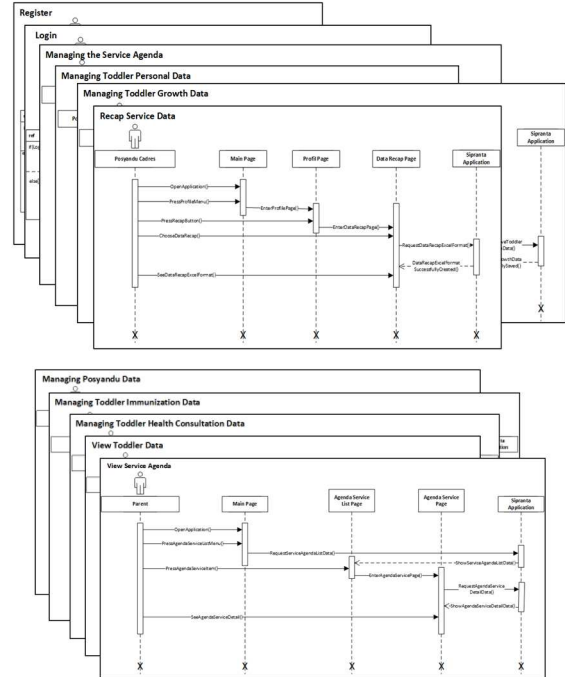


Fig. 4. Snapshot of Sequence Diagram on the Sipranta Application SRS

Sequence Diagrams that are identified are the objects. An example of a Sequence Diagram can be seen in Fig. 4. Then, for this paper, four examples of identification results of all objects are shown in Table V.

TABLE V. IDENTIFICATION RESULTS OF SEQUENCE DIAGRAM

| Document Label | Identifications Results                                                                         |
|----------------|-------------------------------------------------------------------------------------------------|
| d12            | parent, health worker, posyandu cadres. register page. sipranta application.                    |
| d13            | parent, health worker, posyandu cadres. login page. sipranta application. mainpage.             |
| d14            | posyandu cadres. main page.agenda service page. agenda service form page. sipranta application. |
| ...            | ...                                                                                             |
| d22            | parent.main page. agenda service list page. agenda service page. sipranta application.          |

Before the measurement stage, the dataset built previously needs to be Text Pre-processing to maximize the measurement results. In this research, the Natural Language Toolkit (NLTK) is used for text pre-processing. NLTK is a Natural Language Processing (NLP) tool written in the Python-3 computer language. Text Pre-processing, in general, consists of several stages, namely:

- Case Folding** serves to convert letters in the dataset into lowercase letters.
- Tokenization** serves to convert sentences into words.
- Stopword Removal** serves to eliminate connecting words in the dataset.
- Stemming** serves to convert a word into its basic word.

For the Case Folding process, the extraction result from the Step Performed is stored in a document named 'sp\_forecast';

then the program will read the document and produce a document named 'sp\_aftercast,' which is the same document as 'sp\_forecast' but in all lowercase letters. This process is also done in the Sequence Diagram. In the 'sp\_aftercase' document, there is a set of sentences from each Use Case in both artifacts. Each sentence from the Use Case will be identified as a variable called: d1, d2, d3, ..., d22. As can be seen in Fig. 5.

```
#STEP PERFORMED Document
d1 = "parents of toddlers, posyandu cadres, health workers open t
d2 = "parents of toddlers, posyandu cadres, health workers open t
d3 = "posyandu cadres open the sipranta application. posyandu cadr
d4 = "posyandu cadres open the application. posyandu cadres press
d5 = "posyandu cadres open the application. posyandu cadres press
d6 = "posyandu cadres open the sipranta application. posyandu cadres
d7 = "posyandu cadres open the sipranta application. posyandu cadr
d8 = "the health worker opens the sipranta application. the health
d9 = "the health worker opens the sipranta application. health wor
d10 = "parents of toddlers open the sipranta application. parents
d11 = "toddler parents open the sipranta application. parents of t
#Sequence Document
d12 = "parent, health worker, posyandu cadres. register page. sipra
d13 = "parent, health worker, posyandu cadres. login page. sipranta
d14 = "posyandu cadres. main page. agenda service page. agenda servi
d15 = "posyandu cadres. main page. toddler page. toddler form page.
d16 = "posyandu cadres. main page. agenda service page. growth page
d17 = "posyandu cadres. main page. profil page. data recap page. si
d18 = "posyandu cadres. main page. posyandu page. posyandu form pag
d19 = "health workers. main page. toddler list page. toddler page.
d20 = "health workers. main page. consultation page. consultation f
d21 = "parent. main page. toddler list page. toddler page. sipranta
d22 = "parent. main page. agenda service list page. agenda service p

STEP_dataset = [d1,d2,d3,d4,d5,d6,d7,d8,d9,d10,d11]
SEQUENCE_dataset = [d12,d13,d14,d15,d16,d17,d18,d19,d20,d21,d22]
```

Fig. 5. Document Variable

After the document has been identified as a variable, the following process is to convert the sentence into words (Tokenization) and remove the connecting words (Stopword Removal). By removing the connecting words, the measurement results can be maximized. The output of the Tokenization and Stopword Removal method is shown in Fig. 6.

```
{'parent': 17, '': 0, 'health': 9, 'worker': 26, 'posyandu': 19, 'cadres': 4, '': 1, 'register': 22, 'page': 15, 'sipranta': 24, 'application': 3, 'login': 12, 'mainpage': 14, 'main': 13, 'page.a genda': 16, 'service': 23, 'agenda': 2, 'form': 7, 'toddler': 25, 'growth': 8, 'profil': 20, 'dat a': 6, 'recap': 21, 'workers': 27, 'list': 11, 'immunization': 10, 'consultation': 5, 'parent.mai n': 18}
```

Fig. 6. Tokenization and Stopword Removal Process Output

The next stage is to convert each word from the dataset into essential words (Stemming). This process also improves the measurement results if different words have the same semantics. Can be seen in Fig. 7 for the Stemming process.

```
parent of toddler , posyandu cadr , healt
h worker open the sipranta applic . paren
t of toddler , posyandu cadr , health wor
ker press the regist button on the login
page . parent of toddler , posyandu cadr
, health worker input regist data on the
regist page . the sipranta applic valid t
he regist data . the sipranta applic send
a verif email to parent of toddler , posy
andu cadr , health worker . parent of tod
```

Fig. 7. Stemming Process Output

## B. Similarity Measurement

After the Text Analysis process, the next step is conducting the suitability measurement process. The suitability measurement process consists of 2 stages, namely:

- TF-IDF Matrix** serves to weigh each word resulting from Stemming.

- Cosine Similarity** is used to calculate the compatibility of texts based on the TF-IDF matrix results.

TF Matrix is obtained by transforming the previous Stemming results. After that, by using the '.shape' operator from Python, we can find the size of the TF Matrix, which is 11x93. 11x93 here means that from the output, there are 11 documents with the number of indexed words which is 93.

After the results of the TF Matrix have been obtained, the next step is to calculate the IDF Matrix based on the TF Matrix. This process uses the "TfidfTransformer" feature from the Python library "sklearn".

After the TF Matrix and IDF Matrix have been obtained, the next step is calculating the TF-IDF Matrix. TF-IDF Matrix can be obtained by multiplying TF and IDF, where TF Matrix is "11 × 93" and IDF Matrix is "93 × 93". After that TF-IDF Matrix will be divided by Euclidean Norm. The result can be seen in Fig. 8.

|             |            |            |     |    |          |
|-------------|------------|------------|-----|----|----------|
| [0.76921546 | 0.19340836 | 0.         | ... | 0. | 0.302276 |
| [0.63580397 | 0.26643987 | 0.         | ... | 0. | 0.249849 |
| [0.         | 0.37647706 | 0.79007939 | ... | 0. | 0.       |
| ...         |            |            |     |    |          |
| [0.         | 0.333943   | 0.         | ... | 0. | 0.227745 |
| [0.         | 0.29273008 | 0.         | ... | 0. | 0.       |
| [0.         | 0.2651933  | 0.77915308 | ... | 0. | 0.       |

Fig. 8. TF-IDF Matrix calculation result

The following process is to perform measurements using the Cosine Similarity model by obtaining the TF-IDF Matrix values of the two artifacts. The measurement is done by comparing each sentence with all other sentences. Then to avoid the measurement value being too high, the result will be reduced by 0.1 so that the maximum suitability value is 0.9. The results of the suitability between documents are represented in the form of a two-dimensional matrix of 11x11. Can be seen in Fig. 9.

|    |          |          |          |          |          |          |          |
|----|----------|----------|----------|----------|----------|----------|----------|
| 0  | 0.833461 | 0.737086 | 0.184776 | 0.263198 | 0.166393 | 0.071768 | 0.265218 |
| 1  | 0.768257 | 0.838876 | 0.196156 | 0.279905 | 0.177444 | 0.076593 | 0.259881 |
| 2  | 0.187776 | 0.195946 | 0.684721 | 0.471284 | 0.385672 | 0.161879 | 0.418924 |
| 3  | 0.216562 | 0.286166 | 0.337758 | 0.814880 | 0.489122 | 0.160642 | 0.388353 |
| 4  | 0.087241 | 0.172437 | 0.470967 | 0.428609 | 0.729228 | 0.141848 | 0.340024 |
| 5  | 0.123776 | 0.237614 | 0.376565 | 0.511458 | 0.362169 | 0.680916 | 0.432241 |
| 6  | 0.190711 | 0.267014 | 0.473994 | 0.624735 | 0.450149 | 0.238157 | 0.710307 |
| 7  | 0.256054 | 0.309351 | 0.231089 | 0.616897 | 0.351674 | 0.111274 | 0.306929 |
| 8  | 0.248342 | 0.328194 | 0.321185 | 0.459628 | 0.292264 | 0.152769 | 0.275642 |
| 9  | 0.163333 | 0.221188 | 0.285309 | 0.676895 | 0.373706 | 0.092543 | 0.254885 |
| 10 | 0.044777 | 0.124234 | 0.653520 | 0.313729 | 0.309148 | 0.092543 | 0.254885 |

Fig. 9. Measurement results between documents

The results of the conformance measurement between the two artifacts with the same Use Case have the same index in the column and row. Can be seen in Fig. 10 and Fig 11.

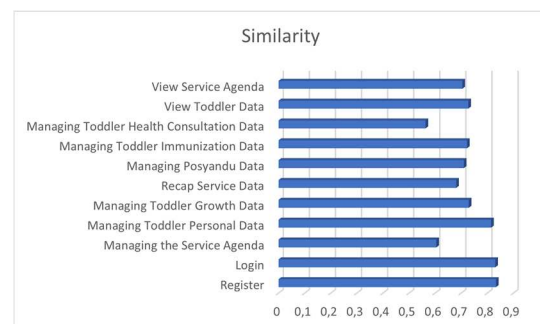


Fig. 10. Chart of measurement results between documents

|      | SD01         | SD02         | SD03         | SD04         | SD05         | SD06         | SD07         | SD08         | SD09         | SD10         | SD11         |
|------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| SP01 | 0.833<br>461 | 0.737<br>006 | 0.184<br>776 | 0.263<br>198 | 0.166<br>393 | 0.071<br>768 | 0.265<br>218 | 0.358<br>250 | 0.373<br>389 | 0.187<br>839 | 0.160<br>609 |
| SP02 | 0.768<br>267 | 0.830<br>876 | 0.196<br>156 | 0.279<br>905 | 0.177<br>444 | 0.076<br>963 | 0.259<br>881 | 0.348<br>189 | 0.362<br>467 | 0.188<br>874 | 0.161<br>546 |
| SP03 | 0.107<br>776 | 0.195<br>946 | 0.604<br>721 | 0.471<br>204 | 0.385<br>672 | 0.161<br>879 | 0.410<br>924 | 0.251<br>648 | 0.300<br>936 | 0.202<br>035 | 0.432<br>645 |
| SP04 | 0.216<br>562 | 0.286<br>166 | 0.337<br>758 | 0.814<br>880 | 0.489<br>122 | 0.160<br>642 | 0.388<br>353 | 0.529<br>955 | 0.631<br>674 | 0.628<br>186 | 0.298<br>318 |
| SP05 | 0.087<br>241 | 0.172<br>437 | 0.470<br>967 | 0.428<br>609 | 0.729<br>220 | 0.141<br>848 | 0.340<br>024 | 0.252<br>337 | 0.294<br>985 | 0.203<br>864 | 0.342<br>980 |
| SP06 | 0.123<br>776 | 0.237<br>614 | 0.376<br>565 | 0.511<br>458 | 0.362<br>169 | 0.680<br>916 | 0.432<br>241 | 0.366<br>757 | 0.345<br>788 | 0.231<br>535 | 0.179<br>730 |
| SP07 | 0.190<br>711 | 0.267<br>014 | 0.473<br>994 | 0.624<br>735 | 0.450<br>149 | 0.238<br>157 | 0.710<br>307 | 0.273<br>201 | 0.321<br>176 | 0.221<br>343 | 0.190<br>944 |
| SP08 | 0.256<br>054 | 0.309<br>351 | 0.231<br>089 | 0.616<br>897 | 0.351<br>674 | 0.111<br>274 | 0.306<br>929 | 0.722<br>319 | 0.765<br>633 | 0.692<br>212 | 0.413<br>237 |
| SP09 | 0.248<br>342 | 0.328<br>194 | 0.321<br>185 | 0.459<br>628 | 0.292<br>264 | 0.152<br>769 | 0.275<br>642 | 0.554<br>207 | 0.563<br>496 | 0.256<br>268 | 0.222<br>565 |
| SP10 | 0.163<br>333 | 0.221<br>188 | 0.205<br>309 | 0.676<br>895 | 0.373<br>706 | 0.092<br>543 | 0.254<br>885 | 0.569<br>871 | 0.656<br>782 | 0.727<br>562 | 0.374<br>751 |
| SP11 | 0.044<br>777 | 0.124<br>234 | 0.653<br>520 | 0.313<br>279 | 0.309<br>148 | 0.092<br>543 | 0.254<br>885 | 0.293<br>757 | 0.323<br>359 | 0.294<br>411 | 0.704<br>504 |

Fig. 11. Measurement results of conformity between Step Performed and Sequence Diagram

### C. Validity & Reliability

This section will use the measurement findings acquired from Cosine Similarity to determine validity and reliability. Gwet's AC1 approach is used to assess validity and reliability. Validity is done in two ways: using the 'cohen\_cappa\_score' feature from the Python 'sklearn.metrics' library and expert judgment. 'sklearn\_metrics' calculates the cohen kappa score using the formula

$$\text{kappa} = \frac{P_o - P_e}{1 - P_e} \quad (6)$$

Information:

$P_o$  is the proportion of agreement between two labels

$P_e$  is the proportion of agreement expected if the agreement is between two labels.

The Cohen Kappa Score is determined by selecting the document with the greatest conformance value from the measurement results. In this study, document 1 (SP01) and document 12 (SD01) are the highest suitability values. Then, the Cohen Kappa Score is obtained, and the value is 0.29195. Can be seen in Fig. 12.

```
cohen_kappa_score(tf_matrixSEQUENCE[0], tf_matrixSTEP[0])
0.2919540229885057
```

Fig. 12. Cohen Kappa Score of measurement results d1 and d12

Furthermore, validity is measured using the opinions of two experts in the SRS document through a questionnaire. Table IV shows the questionnaire responses from two experts.

TABLE VI. HASIL KUESIONER ANTARA DUA AHLI

| Expert 1 | Expert 2 |    |       |
|----------|----------|----|-------|
|          | Yes      | No | Total |
| Yes      | 46       | 7  | 53    |
| No       | 1        | 1  | 2     |
| Total    | 47       | 8  | 55    |

The results of the two experts' questionnaires will be calculated to obtain the validation results. The steps are as follows:

- Calculating P value

$$P = \frac{46+1}{55} = 0.85 \quad (7)$$

- Calculating Probability Change-agreement or  $e(Y)$

$$P_1 = \frac{(47+53)}{55} = 0.9091 \quad (8)$$

$$e(Y) = 2(0.9091)(1-0.9091) = 0.1653 \quad (9)$$

- Calculating validity value or AC1 value

$$AC1 = \frac{0.8 - 0.1653}{1 - 0.1653} = 0.8275 \quad (10)$$

The result of the validation value obtained using the opinion method of two experts is **0.8275**. For documents with a suitability value below the threshold of **0.8** for example (d3, d14), (d5, d16), (d6, d17), improvements can be made by identifying objects that will be used in the Sequence Diagram based on the Step Performed.

## VI. CONCLUSION AND FUTURE WORK

After the measurement data and validation values have been acquired, several conclusions can be drawn. Text Analysis is done on the extracted documents from the Step Performed and the Sequence Diagram, which total 22 documents, 11 from Step Performed and 11 from Sequence Diagram. The Text Analysis procedure produces two papers with the highest conformity, namely:

- d1 = "parents of toddlers, posyandu cadres, health workers open the sipranta application. parents of toddlers, posyandu cadres, health workers press the register button on the login page. parents of toddlers, posyandu cadres, health workers input register data on the register page. the sipranta application validates the register data. the sipranta application sends a verification email to parents of toddlers, posyandu cadres, health workers. parents of toddlers, posyandu cadres, health workers verify accounts."
- d12 = "parent, health worker, posyandu cadres. register page. sipranta application."

Measurement results of d1 and d12 using the Cosine Similarity model by obtaining a suitability value of **0.833461**.

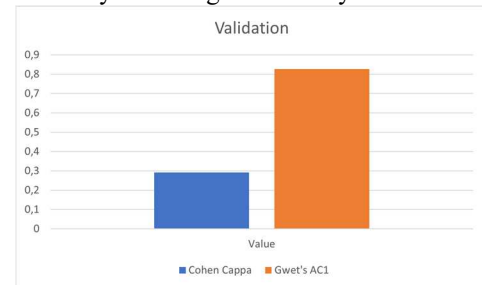


Fig. 13. Comparison between two validation methods

Validation of the suitability measurement results is done by two methods, namely using the 'cohen\_cappa\_score' tool from the 'sklearn.metrics' python library and the opinions of 2 experts. As shown in Fig. 13, the 'cohen\_cappa\_score' produces a validation value of **0.29195** which is included in the "Fair Agreement" category, while the validation results from 2 experts are **0.8275**, which is included in the "Substantial

agreement" category. There is a difference in results between the two methods in the amount of **0.536**.

This study measures the Step Performed and Sequence Diagram conformity by implementing Text Pre-processing. Has been achieved with a conformity value of **0.833461** which means that the two documents are in accordance. The validation results in fall into the category of "Fair Agreement" using Cohen's Cappa and "Substantial Agreement" using Gwet's AC1, where these results are slightly different from the results of previous measurements. Then the measurement results can be considered valid and reliable, and there is conformity between the two artifacts on the SRS Sipranta Application. For documents with a suitability value below the thresh hold of **0.8** for example (d3, d14), (d5, d16), (d6, d17), improvements can be made by identifying objects that will be used in the Sequence Diagram based on the Step Performed.

For future work, several things can be developed, including the use of larger datasets to strengthen the quality of measurements and the application of more sophisticated natural language management techniques; this research uses NLTK from python-3, there are more sophisticated natural languages management technologies such as Spacy, Transformers, and GPT3, and the expansion of research to different domains such as information extraction from medical or legal documents.

#### ACKNOWLEDGMENT

#### REFERENCES

- [1] Society, IEEE Computer, "IEEE Recommended Practice for Software Requirements Specifications," *Software Engineering Standards Committee*, p. 39, 2011.
- [2] Iyoda, Y., Ohnishi, A., " Software requirements parts for construction of software requirements specifications," ICSoft 2013 - Proceedings of the 8th International Joint Conference on Software Technologies pp. 147-153, 2013.
- [3] Zhao, L., Alhoshan, W., Ferrari, A., Letsholo, K. J., Ajagbe, M. A., Chioasca, E. V., & Batista-Navarro, R. T., "Natural language processing for requirements engineering: A systematic mapping study". *ACM Computing Surveys (CSUR)*, 2021.
- [4] R. P. Octavially, Y. Priyadi and S. Widowati, "Extraction of Activity Diagrams Based on Steps Performed in Use Case Description Using Text Mining (Case Study: SRS Myoffice Application)," in *International Conference on Electrical and Electronic Intelegant System (ICE3IS)*, Yogyakarta, 2022.
- [5] Ali Olow Jim'ale Sabriye and Wan Mohd Nazmee Wan Zainon, "A Framework For Detecting Ambiguity In Software," in *International Conference on Information Technology (ICIT)*, Shanghai, 2017.
- [6] M. Asif, I. Ali, M. S. A. Malik, M. H. Chaudary, S. Tayyaba and M. T. Mahmood, "Annotation of Software Requirements Specification (SRS), Extractions of Nonfunctional Requirements, and Measurement of Their Tradeoff," *IEEE Access*, p. 13, 2019.
- [7] Zhou, H., Huang, Z., Wang, L., " UML sequence diagram metrics based on polymorphism," *Nanjing Hangkong Hangtian Daxue Xuebao/Journal of Nanjing University of Aeronautics and Astronautics* 38(6), pp. 759-763, 2006.
- [8] F. Hujainah, R. B. A. Bakar, M. Abdullateef, Abdulgaber and K. Z. Zamli, "Software Requirements Prioritisation: A Systematic Literature Review on Significance, Stakeholders, Techniques and Challenges," *IEEE Access*, vol. XX, p. 29, 2017.
- [9] Y. Priyadi, A. M. Putra and P. S. Lyanda, "The similarity of Elicitation Software Requirements Specification in Student Learning Applications of SMKN7 Baleendah Based on Use Case Diagrams Using Text Mining," in *2021 IEEE 5th International Conference on Information Technology, Information Systems and Electrical Engineering (ICITISEE)*., Yogyakarta, 2021.
- [10] Friske, M., Schlingloff, B.-H., " Generation of UML models from formalized use case descriptions," Tagungsband - Dagstuhl-Workshop MBES: Modellbasierte Entwicklung eingebetteter Systeme III, MBES 2007 pp. 113-121, 20207.
- [11] S. K. Swain, D. P. Mohaputra and R. Mall, "Test Case Generation Based on Use case and Sequence Diagram," *International Journal of Software Engineering*, p. 33, 2010.
- [12] C. Alvin, B. Peterson and S. Mukhopadhyay, "Static generation of UML sequence diagrams," *International Journal on Software Tools for Technology Transfer*, 2021.
- [13] Jo, T., "Text mining", *Studies in Big Data*, 2019.
- [14] V. Gupta and G. S. Lehal, "A Survey of Text Mining Techniques and Applications," *Journal of emerging technologies in web intelligence*, vol. I, p. 17, 2019.
- [15] Y. Priyadi, K. Kusumahad and P. S. Lyanda, "IdVar4CL: Causal Loop Variable Identification Method for Systems Thinking Based on Text Mining Approach," *International Journal of Fuzzy Logic and Intelligent Systems*, 2021.
- [16] S. Kumar, A. K. Kar and P. Vigneswarallavarasan, "Applications of text mining in services management: A systematic literature review," *International Journal of Information Management Data Insights*, p. 14, 2021.
- [17] N. Carey, M. Harte and L. M. Cullagh, "A text-mining tool generated title-abstract screening workload savings:performance evaluation versus single-human screening," *Journal of Clinical Epidemiology*, p. 7, 2022.
- [18] S. Vijayarani, J. Ilamathi and M. Nithya, "Preprocessing Techniques for Text Mining," *International Journal of Computer Science & Communication Networks*, 2014.
- [19] F. Rahutomo, T. Kitasuka and M. Aritsugi, "Semantic Cosine Similarity," in *The 7th international student conference on advanced science and technology ICAST*, 2014.
- [20] Jain, M., Rastogi, H., " Automatic Text Summarization using Soft-Cosine Similarity and Centrality Measures," Proceedings of the 4th International Conference on Electronics, Communication and Aerospace Technology, ICECA 2020 9297583, pp. 1021-1028, 2020.
- [21] R. Samuel, R. Natan and F. Syafiqoh, "Penerapan Cosine Similarity dan K-Nearest Neighbor (K-NN) pada Klasifikasi dan Pencarian Buku," *Journal of Big Data Analytic and Artificial Intelligence*, vol. 1, p. 6, 2018.
- [22] Casey, P., Altobelli, G., Pignatelli, P., " Application of the hypothesis analysis method using Cohen's Kappa index to measure the agreement between leather sorters," 32nd Congress of the International Union of Leather Technologists and Chemist Societies, IULTCS 2013.
- [23] A. R. Lahitani, A. E. Permanasari and N. A. Setiawan, "Cosine similarity to determine similarity measure: Study case in online essay assessment," 2016 4th International Conference on Cyber and IT Service Management, 2016.
- [24] Da Silva, C.F., Osório, F.S., Vieira, R., " Evaluating the use of linguistic information in the pre-processing phase of Text Mining," *Inteligencia Artificial* 9(26), pp. 59-66, 2005.